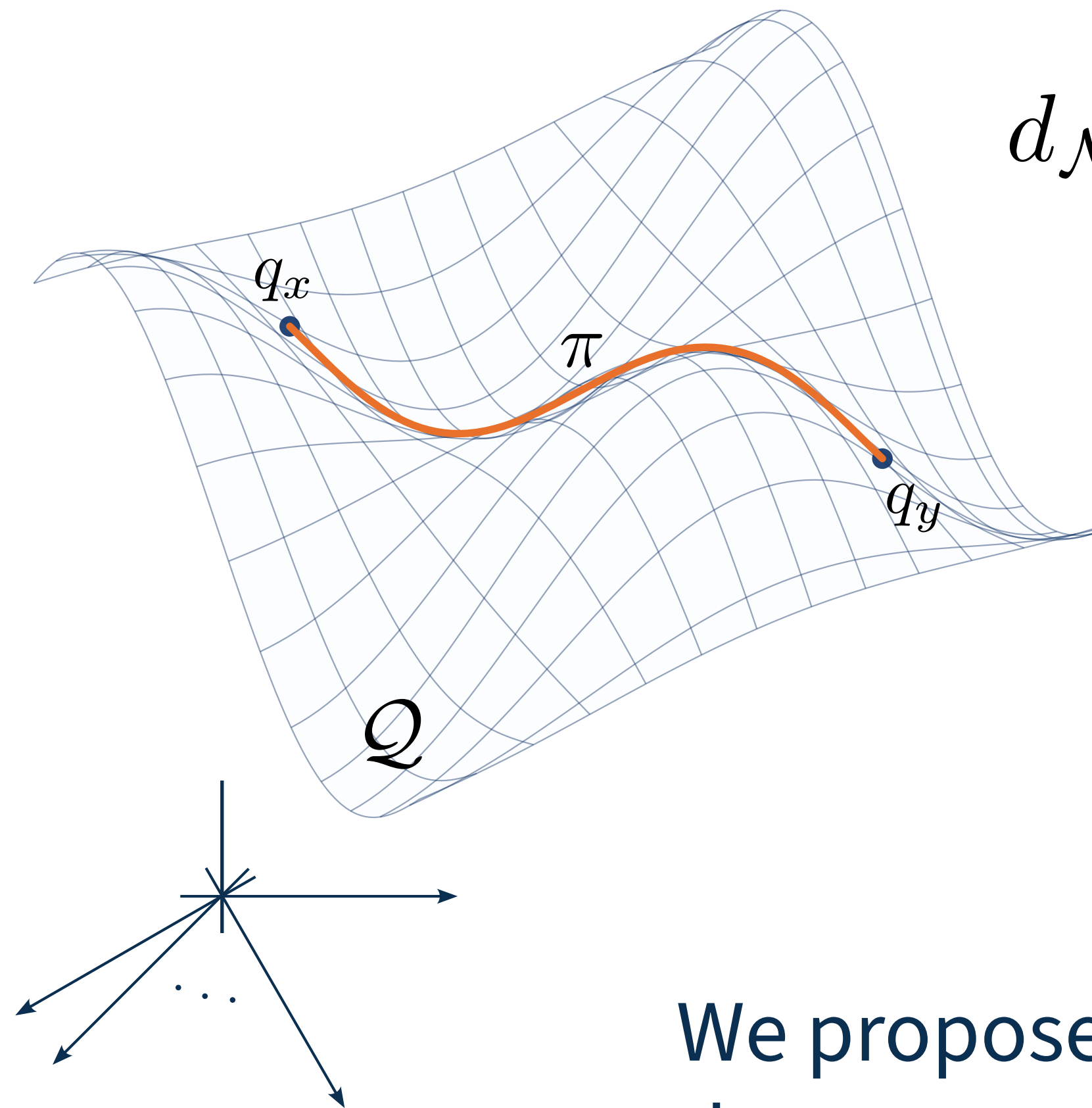


03

Approximating Geodesics

... and can we measure it cheaply?



$$d_{\mathcal{M}}(q_x, q_y) = \min_{\pi} \int_0^1 \sqrt{\dot{\pi}(t)^T \underbrace{G(\pi(t))}_{\text{Riemannian metric}} \dot{\pi}(t)} dt$$

computing this *exactly* requires solving a **two-point boundary-value problem**

We propose a *lightweight* approximation that recovers the **true geodesic distance** when paired with asymptotically optimal planners like RRT*